



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

19CG02647

B.Sc DEGREE EXAMINATION

COLLEGE: SCIENCE AND TECHNOLOGY

SESSION: 2021/2022

COURSE CODE: CSC317

COURSE TITLE: SYSTEM ANALYSIS AND DESIGN

INSTRUCTION: READ THE SCENARIO AND ANSWER **THREE (3) QUESTIONS IN ALL**

DEPARTMENT: Computer & Infor. Sciences

SEMESTER: ALPHA

CREDIT UNIT: 2

TIME: 2 HOURS

A Video Store Scenario

A Video Store (AVS) runs a series of fairly standard video stores. Before a video can be put on the shelf it must be catalogued and entered into the video database. Every customer must have a valid AVS customer card in order to rent a video. Customers rent videos for three days at a time. Every time a customer rents a video, the system must ensure that this customer does not have any overdue videos. If so, the overdue videos must be returned and an overdue fee paid before the customer can rent more videos. Every morning, the store manager prints a report that lists overdue videos; if a video is two or more days overdue, the manager calls the customer to remind him or her to return the video. If a video is returned in damaged condition, the manager removes it from the video database and may sometimes charge the customer.

QUESTION ONE

- Supposing you are a System analyst working for a firm on a system development project. The system has unclear user requirements, a short time schedule and schedule visibility. What methodology will you recommend? Discuss this methodology, stating its advantages and disadvantages. [5marks]
- Let's assume you are a project manager using the waterfall development methodology on an enormous and complex project. Your manager is advocating on replacing the waterfall methodology with rapid application development and has instructed you to switch. What would be your response? [5marks]
- Identify and briefly describe the types of relationships in Use cases [4.5marks]
- Identify the Use cases and sketch a Use Case diagram to depict the video store scenario. [9 marks]

QUESTION TWO

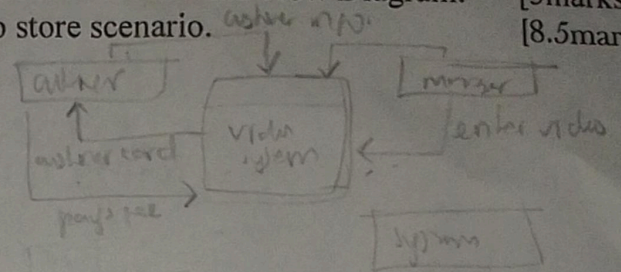
- Differentiate between rapid application development (RAD) and object-oriented design (OOD) methodologies? [5marks]
- Discuss the feasibility aspects used in systems planning, and for each provide an example of associated risk. [5marks]
- i) Differentiate between cardinality and ordinality. ii) describe cardinal relationships that you know. [4.5marks]
- List the entities in the Video store scenario and draw an Entity Relationship Diagram for the scenario (hints: use any notation of your choice and show cardinality where necessary). [9marks]

QUESTION THREE

- What is the System Development Life Cycle (SDLC)? Discuss briefly the major phases in the System Development Life cycle and identify the deliverables in each phase. [10marks]
- Using relevant examples, briefly explain three (3) mechanical errors in Data Flow Diagram. [5marks]
- Draw Process 0 and Level 0 Data Flow Diagrams the video store scenario. [8.5marks]

QUESTION FOUR

- Explain each of the following with an example:
 - a model,
 - methodology,
 - a tool,
 - a technique.
- Identify the overall strategy for fact finding in requirement analysis. [6marks]
- Briefly describe the following Object-Oriented Design concepts with relevant examples:
 - Polymorphism,
 - Inheritance,
 - Composition
 - Aggregation.
- Draw the Class diagram for the video store information system. Show relationships and multiplicities where necessary. [2.5marks]



QUESTION FIVE

- What are the principles of a successful systems development? [5marks]
- How does an analyst solve problems? Discuss. [6marks]
- With the aid of a diagram, briefly identify the required skills of a system's analyst? [3.5marks]
- Name any three (3) Use Cases in the video store scenario and draw sequence diagram for each. [9marks]